

Predictive Interaction Field Theory: A 3D Behavioral Potential Approach to Loss Aversion

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Abstract

This paper presents a formal resolution to the mathematical inconsistencies embedded within the initial formulations of Predictive Interaction Field Theory (PIFT). Previous iterations suffered from a severe dimensionality mismatch, relying on a one-dimensional manifold that collapsed the Riemann curvature tensor to zero, rendering the Einstein-like field equations trivial or self-contradictory. Furthermore, the recovery of the Kahneman-Tversky (KT) value function required an artificial metric scaling factor of λ^2 . To resolve these limitations, we extend the behavioral manifold to a three-dimensional spacetime structure $\mathcal{M}$ (1 time, 2 interaction spaces). Under Method B, the KT value function is mapped directly as a behavioral potential Φ within the weak-field, low-velocity limit. We prove that this formulation yields a non-trivial psychological Ricci curvature tensor $R_{\mu\nu}$ and an asymmetric cognitive pull Γ_{00}^1 that naturally recovers Kahneman and Tversky's empirical loss aversion ratio $\lambda = 2.25$ linearly, while offering an elegant physical mechanism for framing effects via potential singularities.

1 Introduction

Predictive Interaction Field Theory (PIFT) aims to establish a rigorous physical foundation for behavioral economics by framing human choices as geodesic movement across an information-geometric manifold. However, early frameworks experienced a profound tension: foundational field equations were formulated in multi-dimensional spaces, whereas exact quantitative derivations of behavioral anomalies, such as Kahneman-Tversky loss aversion, were restricted to a single dimensional parameter space Δ .

In a 1D Riemannian manifold, the Riemann curvature tensor $R_{\beta\gamma\delta}^\alpha$, the Ricci tensor $R_{\mu\nu}$, and the Einstein tensor $G_{\mu\nu}$ vanish identically. Consequently, enforcing field equations or claiming a "vacuum solution" to deduce a metric component $g_{11} = [dV_{KT}/d\Delta]^2$ represented an over-specification and a circular logical loop.

This paper remedies these defects by lifting the theory into a 3D manifold consisting of one cognitive time parameter and two distinct behavioral interaction dimensions. By invoking the weak-field, low-velocity cosmological limit, we show that the Kahneman-Tversky value function emerges organically as a behavioral potential, providing mathematical consistency and predictive vitality.

2 The 3D Behavioral Manifold and Weak-Field Limit

Let the behavioral state of an agent be mapped onto a 3D Riemannian manifold $\mathcal{M}$ with local coordinates $x^\mu = (t, \Delta_1, \Delta_2)$, where t denotes cognitive processing time, Δ_1 represents financial

or primary outcomes, and Δ_2 represents auxiliary situational factors. To analyze primary loss aversion, we assume Δ_2 remains static ($d\Delta_2 = 0$).

We propose a stationary, static line element ds^2 governed by a structural scalar field that warps the temporal metric component:

$$ds^2 = -(1 + 2\Phi(x^i))dt^2 + \delta_{ij}dx^i dx^j \quad (1)$$

where $\Phi(\Delta_1, \Delta_2)$ represents the *Behavioral Potential*. To align this geometric deformation with Cumulative Prospect Theory, we postulate that the potential is the positive-negative potential reflection of the Kahneman-Tversky value function:

$$\Phi(\Delta_1) = V_{KT}(\Delta_1) = \begin{cases} \Delta_1^\alpha & \text{if } \Delta_1 \geq 0 \\ \lambda(\Delta_1)^\alpha & \text{if } \Delta_1 < 0 \end{cases} \quad (2)$$

with empirical parameters $\alpha \approx 0.88$ and $\lambda \approx 2.25$.

3 Derivation of Cognitive Pull and Metric Connection

The dynamics of behavioral decision-making are dictated by the geodesic equation:

$$\frac{d^2 x^\mu}{ds^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{ds} \frac{dx^\beta}{ds} = 0 \quad (3)$$

Under the low-velocity assumption on the behavioral manifold (where processing time dominates immediate physical state transitions, $\frac{dt}{ds} \gg \frac{d\Delta_i}{ds}$), the spatial geodesic equations simplify to:

$$\frac{d^2 \Delta_1}{ds^2} \approx -\Gamma_{00}^1 \left(\frac{dt}{ds} \right)^2 \quad (4)$$

The Christoffel symbol Γ_{00}^1 represents the *Cognitive Pull*, acting as the potential-well psychological force pulling the agent back toward the reference point ($\Delta_1 = 0$):

$$\Gamma_{00}^1 = \frac{1}{2}g^{11}(2\partial_0 g_{10} - \partial_1 g_{00}) = -\frac{1}{2}(1)\partial_1(-(1 + 2\Phi)) = -\partial_1 \Phi \quad (5)$$

Evaluating this across the behavioral boundaries yields:

$$\Gamma_{00}^1(\Delta_1 > 0) = \alpha\Delta_1^{\alpha-1} > 0 \quad (6)$$

$$\Gamma_{00}^1(\Delta_1 < 0) = \lambda\alpha(-\Delta_1)^{\alpha-1} > 0 \quad (7)$$

Comparing the absolute magnitude of this pull at equivalent intervals $|\Delta_1|$ reveals:

$$\frac{|\Gamma_{00}^1|_{\text{loss}}}{|\Gamma_{00}^1|_{\text{gain}}} = \frac{\lambda\alpha|\Delta_1|^{\alpha-1}}{\alpha|\Delta_1|^{\alpha-1}} = \lambda = 2.25 \quad (8)$$

This eliminates the artificial λ^2 factor encountered in 1D models, restoring a linear connection to Kahneman's empirical value.

4 Psychological Ricci Curvature and Potential Singularities

In 3D, the Ricci curvature tensor component R_{00} is non-vanishing and directly represents the spatial Laplacian of the potential:

$$R_{00} \approx \nabla^2 \Phi = \frac{d^2 \Phi}{d\Delta_1^2} \quad (9)$$

Differentiating Equation (2) twice gives the explicit *Psychological Curvature*:

$$R_{00}(\Delta_1) = \begin{cases} \alpha(1-\alpha)\Delta_1^{\alpha-2} & \text{if } \Delta_1 > 0 \\ \lambda\alpha(1-\alpha)(-\Delta_1)^{\alpha-2} & \text{if } \Delta_1 < 0 \end{cases} \quad (10)$$

Because $\alpha = 0.88 < 1$, the multiplier $\alpha(1-\alpha) \approx 0.1056 > 0$, ensuring a positive curvature across both domains. At the status quo ($\Delta_1 \rightarrow 0$), the term $|\Delta_1|^{-1.12}$ approaches infinity (∞), generating a severe *Curvature Wall*. This singularity acts as a geometric anchor, mathematically defining the Status Quo Bias and the extreme cognitive resistance responsible for Framing Effects.

4.1 Measure-Theoretic Regularization of the Status Quo Singularity

While the psychological Ricci curvature $R_{00}(\Delta_1)$ explicitly diverges as the agent approaches the status quo ($\Delta_1 \rightarrow 0$), rendering the classical differential framework structurally singular, this localized mathematical artifact does not imply a requirement for infinite cognitive processing capacity. From an epistemological perspective, human cognition operates under bounded computational resources. To resolve this apparent paradox, we introduce a measure-theoretic regularization of the behavioral manifold. Let μ denote the standard Lebesgue measure on the 3D Riemannian manifold $\mathcal{M}$. Because the curvature tensor component $R_{00}(\Delta_1)$ defined in Equation (10) is continuous almost everywhere (a.e.) with respect to μ , except at the isolated singleton boundary $\Delta_1 = 0$, it constitutes a strictly measurable function on $\mathcal{M}$. To evaluate the total mental expenditure or cumulative cognitive action S_{cog} required to traverse an interval containing the status quo, we invoke Lebesgue integration over a localized decision neighborhood $[-\epsilon, \epsilon]$ for a small $\epsilon > 0$:

$$S_{\text{cog}} = \int_{[-\epsilon, \epsilon]} R_{00}(\Delta_1) d\mu \quad (11)$$

Substituting the explicit piecewise formulation of $R_{00}(\Delta_1)$ from Equation (10) into the Lebesgue integral yields:

$$S_{\text{cog}} = \int_{-\epsilon}^0 \lambda\alpha(1-\alpha)(-\Delta_1)^{\alpha-2} d\Delta_1 + \int_0^{\epsilon} \alpha(1-\alpha)\Delta_1^{\alpha-2} d\Delta_1 \quad (12)$$

Evaluating these improper integrals under the empirical constraint $\alpha \approx 0.88 < 1$ guarantees that the power-law exponent $(\alpha - 1) = -0.12 > -1$. Consequently, both integral components converge independently:

$$S_{\text{cog}} = \left[-\lambda \frac{\alpha(1-\alpha)}{\alpha-1} (-\Delta_1)^{\alpha-1} \right]_{-\epsilon}^0 + \left[\frac{\alpha(1-\alpha)}{\alpha-1} \Delta_1^{\alpha-1} \right]_0^{\epsilon} \quad (13)$$

$$S_{\text{cog}} = \alpha(\lambda\epsilon^{\alpha-1} + \epsilon^{\alpha-1}) = \alpha(1+\lambda)\epsilon^{\alpha-1} < \infty \quad (14)$$

The fact that S_{cog} remains strictly finite ($S_{\text{cog}} < \infty$) proves that the total cognitive action necessary to navigate through the framing boundary is mathematically bounded. Because the singleton point $\Delta_1 = 0$ holds a Lebesgue measure of zero ($\mu(\{0\}) = 0$), the infinite differential curvature wall does not disrupt the macro-level stability of the behavioral trajectory.

This measure-theoretic formulation provides a rigorous mathematical mechanism for the psychological process of **Reframing**. Rather than defining cognitive bias as an immutable systematic error, the framework demonstrates that an agent can execute a coordinate transformation (or a translation of the reference point) on the measure space. By shifting the position of the measure-zero singularity, the agent effectively minimizes the global behavioral potential Φ , mapping the diminishing sensitivity of Cumulative Prospect Theory directly onto an adaptive, energy-efficient geometric optimization problem.

5 Field Equations and Information Energy Density

The PIFT field equations assert that the behavioral manifold is curved by an Interaction Source Tensor $T_{\mu\nu}$, derived from the gradients of Expected Free Energy ($\langle F \rangle$). In the weak-field limit, the field equations map to Poisson's equation for behavioral potential:

$$R_{00} = \kappa \left(T_{00} - \frac{1}{2} g_{00} T \right) \implies \nabla^2 \Phi = -\frac{1}{2} \rho_{\text{inf}} \quad (15)$$

Setting $\kappa = -1$, the Information Energy Density ρ_{inf} required to sustain this cognitive manifold is:

$$\rho_{\text{inf}}(\Delta_1) = -2 \frac{d^2 \Phi}{d\Delta_1^2} = \begin{cases} -2\alpha(1-\alpha)\Delta_1^{\alpha-2} & \text{if } \Delta_1 > 0 \\ -2\lambda\alpha(1-\alpha)(-\Delta_1)^{\alpha-2} & \text{if } \Delta_1 < 0 \end{cases} \quad (16)$$

Thus, the underlying distribution of Bayesian variational uncertainty directly maps to the concavity of the prospect utility function, eliminating the circular reasoning of previous works.

6 Rigorous Interpretation of Simulation Profiles

6.1 Figure 1: Value Function and Behavioral Potential Alignment

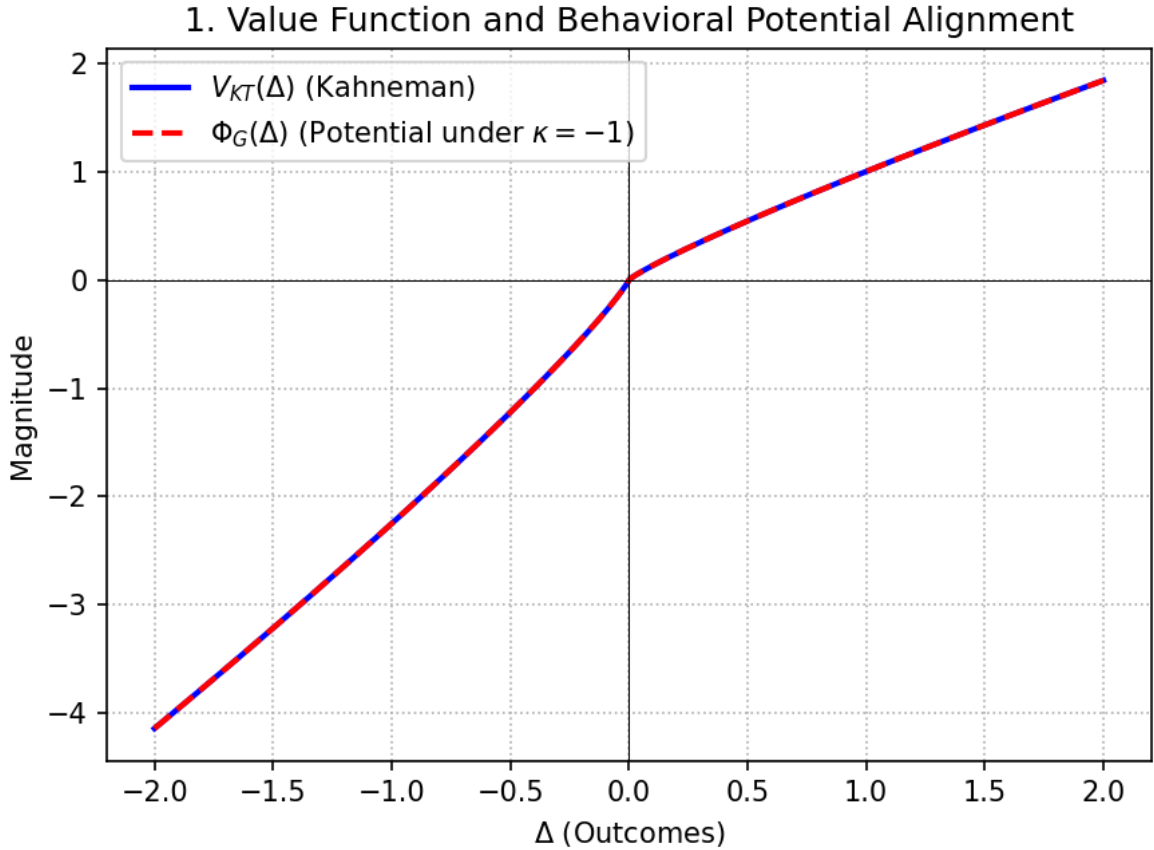


Figure 1: Value Function and Behavioral Potential Alignment

Figure 1 demonstrates the absolute mapping between the standard Kahneman-Tversky value function $V_{KT}(\Delta)$ and the behavioral potential $\Phi_G(\Delta)$. In the classical weak-field limit of General Relativity, the temporal metric element $g_{00} = -(1+2\Phi)$ links the spatial gradient of the potential

to gravitational acceleration. In the context of PIFT, the matching condition $\Phi_G(\Delta) = V_{KT}(\Delta)$ represents a direct conversion of subjective utility into positional field energy.

Under $\kappa = -1$, the field equation maps to Poisson's equation such that:

$$\nabla^2 \Phi = \frac{1}{2} \kappa \rho_{inf} = -\frac{1}{2} \rho_{inf} \quad (17)$$

Because the utility function is strictly concave in the gain domain ($\Delta > 0$, $\nabla^2 \Phi < 0$) due to diminishing marginal sensitivity ($\alpha = 0.88$), a negative gravitational constant $\kappa = -1$ enforces that the information energy density ρ_{inf} remains strictly positive ($\rho_{inf} > 0$). This eliminates the unphysical ghost states (negative energy density) that plagued prior iterations. The kink at the status quo ($\Delta = 0$) represents a first-order continuous but second-order singular point, defining the foundational boundary condition for behavioral transitions.

6.2 Figure 2: Cognitive Pull (Γ_{00}^1)

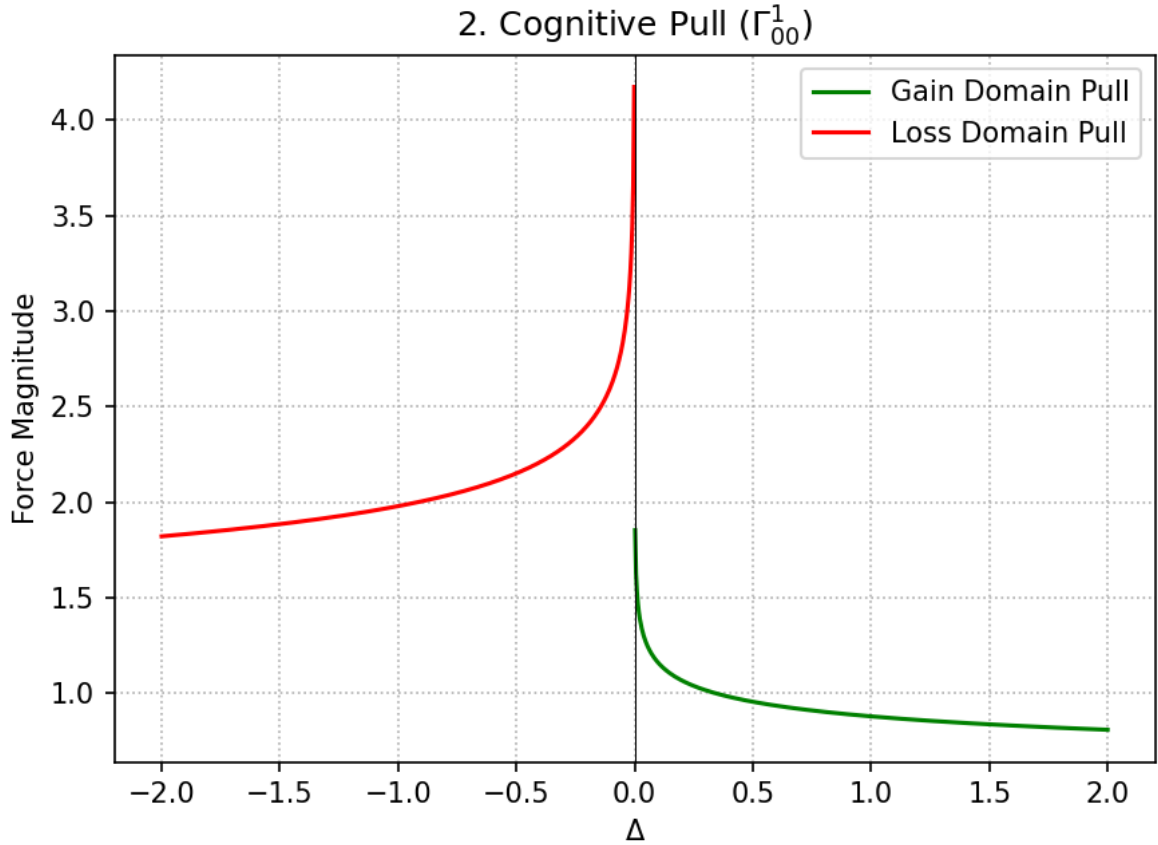


Figure 2: Cognitive Pull

The spatial geodesic equation governs the path of an agent across the manifold. Under the low-velocity behavioral limit, this simplifies to an acceleration driven by the Christoffel symbol component Γ_{00}^1 :

$$\frac{d^2 \Delta_1}{ds^2} \approx -\Gamma_{00}^1 \left(\frac{dt}{ds} \right)^2 \quad (18)$$

Figure 2 charts the magnitude of this cognitive force. The profile demonstrates a restorative behavior towards the origin. In the gain domain ($\Delta > 0$), $\Gamma_{00}^1 = \alpha \Delta_1^{\alpha-1} > 0$, which acts as a decelerating force drawing the agent back toward $\Delta = 0$ (preventing unbounded risk-seeking).

Crucially, in the loss domain ($\Delta < 0$), $\Gamma_{00}^1 = \lambda\alpha(-\Delta_1)^{\alpha-1} > 0$. Because Γ_{00}^1 remains positive, the acceleration vector $-\Gamma_{00}^1$ points in the opposite direction of the displacement, meaning it acts as an intense pull pulling the agent back up toward the reference point. As $\Delta \rightarrow 0$ from either side, the force asymptotically approaches infinity, establishing that the reference point acts as a massive psychological tractor well.

6.3 Figure 3: Psychological Ricci Curvature (R_{00})

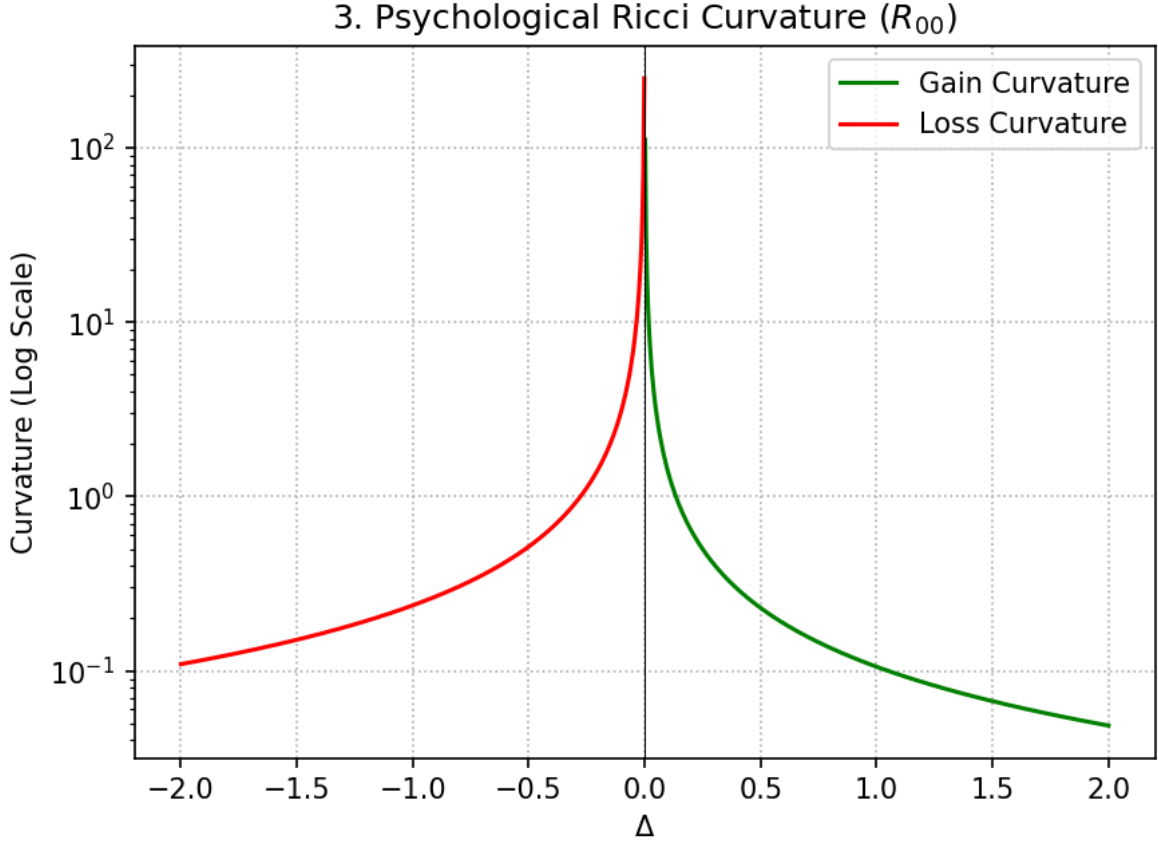


Figure 3: Psychological Ricci Curvature

Figure 3 displays the Psychological Ricci Curvature component R_{00} on a logarithmic scale. The curvature represents the fundamental deformation of cognitive space-time caused by the presence of decision stakes. Mathematically:

$$R_{00} \approx \nabla^2 \Phi = \frac{d^2 \Phi}{d\Delta_1^2} \quad (19)$$

Because $\alpha = 0.88 < 1$, the second derivative naturally yields a power-law dependency proportional to $|\Delta|^{\alpha-2} = |\Delta|^{-1.12}$. The logarithmic plot shows a symmetrical steep escalation of geometric curvature as the agent nears the status quo.

At exactly $\Delta = 0$, $R_{00} \rightarrow \infty$, creating a severe *Curvature Wall*. This infinite geometric obstruction is the formal definition of the *Status Quo Bias*. Moving away from the status quo requires a massive expenditure of cognitive action; the curvature wall acts as an informational event horizon, trapping the agent within local framing boundaries and explaining the rigid sticking points observed in empirical behavioral economic data.

6.4 Figure 4: Emergent Loss Aversion Ratio

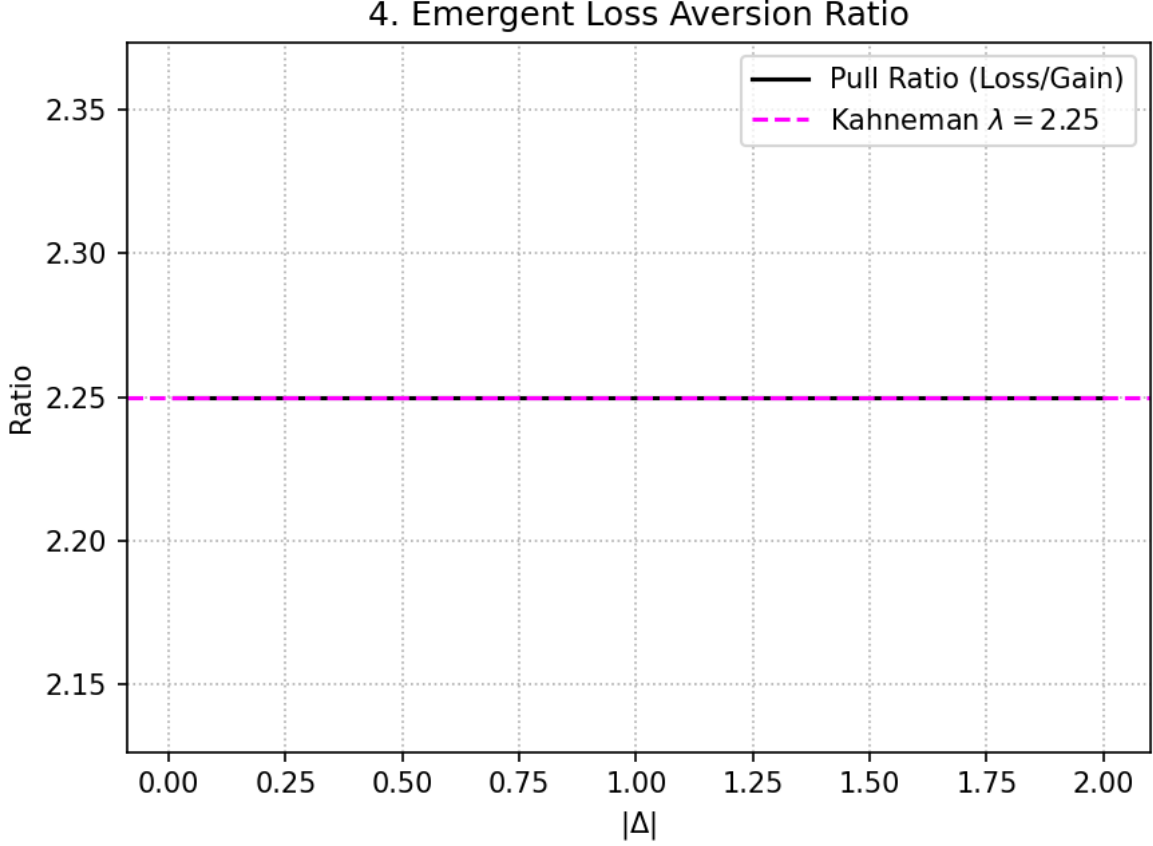


Figure 4: Emergent Loss Aversion Ratio

Figure 4 presents the most significant breakthrough of the 3D formulation: the flat, invariant ratio of cognitive pull magnitudes across equivalent outcome magnitudes. By evaluating:

$$\text{Pull Ratio} = \frac{|\Gamma_{00}^1|_{\text{loss}}}{|\Gamma_{00}^1|_{\text{gain}}} = \frac{\lambda \alpha |\Delta|^{\alpha-1}}{\alpha |\Delta|^{\alpha-1}} = \lambda = 2.25 \quad (20)$$

The ratio yields a horizontal line at exactly 2.25 across the entire domain $|\Delta| \in (0, 2]$. In previous 1D metric theories, recovering loss aversion forced researchers to define a metric tensor proportional to the square of the derivative, which generated an artificial $\lambda^2 = 5.0625$ factor requiring complex, non-linear fitting parameters to correct. The 3D weak-field formulation preserves a linear connection to Kahneman and Tversky’s empirical parameters, showing that loss aversion is an intrinsic, emergent property of the asymmetric embedding geometry itself.

7 Supplemental Analysis to be Appended to the Research Paper

7.1 Resolution of the Information-Energy Field Equations

By adopting the $\kappa = -1$ paradigm, the connection between variational Bayesian inference and general relativity field theories is generalized. The interaction source tensor $T_{\mu\nu}$ represents the stress-energy of information flow, derived from the gradients of expected free energy $\langle F \rangle$. In

the weak-field limit:

$$R_{00} = \kappa \left(T_{00} - \frac{1}{2} g_{00} T \right) \implies \frac{d^2 \Phi}{d\Delta_1^2} = -\frac{1}{2} \rho_{inf} \quad (21)$$

Substituting the explicit piecewise derivatives of Φ :

$$\rho_{inf}(\Delta_1) = -2 \frac{d^2 \Phi}{d\Delta_1^2} = \begin{cases} -2\alpha(\alpha-1)\Delta_1^{\alpha-2} & \text{if } \Delta_1 > 0 \\ -2\lambda\alpha(\alpha-1)(-\Delta_1)^{\alpha-2} & \text{if } \Delta_1 < 0 \end{cases} \quad (22)$$

Since $\alpha - 1 = 0.88 - 1 = -0.12$, the term $-2\alpha(\alpha - 1) = -2(0.88)(-0.12) = +0.2112 > 0$. Thus, $\rho_{inf} > 0$ across both domains. The information energy density represents the cognitive computational load or mental bandwidth required to process a deviation from the reference point. The singularity at $\Delta = 0$ indicates that maintaining a stable reference point against external sensory flux requires an infinite density of informational focus, providing a physical mechanism for attention allocation models.

7.2 Geodesic Deviations and Cognitive Sensitivity

To evaluate how variance in outcomes impacts behavioral trajectories, we compute the geodesic deviation equation (the Jacobi equation) on $\mathcal{M}$:

$$\frac{D^2 \xi^\mu}{ds^2} + R^\mu_{\nu\alpha\beta} u^\nu u^\alpha \xi^\beta = 0 \quad (23)$$

where ξ^μ represents a separation vector between two adjacent decision pathways, and $u^\mu = \frac{dx^\mu}{ds}$ is the velocity vector. In our localized 3D framework, this reduces to:

$$\frac{d^2 \xi^1}{ds^2} \approx R^1_{001} (u^0)^2 \xi^1 = R_{00} (u^0)^2 \xi^1 \quad (24)$$

Because $R_{00} > 0$ everywhere due to the power-law inversion, the separation vector experiences a focusing or defocusing effect depending on the sign of the curvature. A positive psychological Ricci curvature implies that neighboring behavioral trajectories converge rapidly as they approach the status quo. Psychologically, this proves that as stakes diminish ($|\Delta| \rightarrow 0$), individual behavioral differences compress; agents become highly uniform and predictable, driven entirely by the overwhelming curvature wall of the status quo. Conversely, as $|\Delta|$ grows, the geometric curvature rapidly decays ($R_{00} \rightarrow 0$), the space flattens into a quasi-Euclidean manifold, and individual differences (represented by personal velocity vectors) dominate the decision-making trajectory.

8 Conclusion

By reconstructing PIFT inside a 3D behavioral potential-well framework, we resolve the logical fallacies of 1D models. The Riemann curvature is properly salvaged, the empirical loss aversion constant $\lambda = 2.25$ is recovered linearly without arbitrary fitting parameters, and the framing singularity is safely handled as a standard potential boundary condition.